
What Makes the HVI Color Space Work?

A Controlled Study of Dual-Space Losses, Width, and Input Prefiltering in HVI-CIDNet

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Abstract

HVI-CIDNet is a low-light image enhancement network that is trained with a loss applied in two color spaces at once: the standard RGB space and the HVI (Horizontal/Vertical/Intensity) space it introduces. Which parts of that recipe actually matter is not established. We run a controlled study of the HVI-CIDNet training recipe on LOL-v1 under a single frozen protocol: (1) a leave-one-out ablation of the eight loss terms (L1, SSIM, edge, and perceptual, each in RGB and in HVI), (2) model width from 18 to 36 base channels, and (3) intensity-only and hue-value input prefilterers. All runs share the same split, seeds, optimizer, schedule, and hardware (a single NVIDIA GeForce RTX 2080 Ti (11 GB) in FP32), and every reported number is read from a result file rather than transcribed. We find that removing the HVI-space perceptual term changes validation PSNR by 0.3 dB and LPIPS by 0.02, that widening from 18 to 36 channels gains 0.6 dB at 4 $\times$ the parameter count, and that the intensity-only prefilter does not reproduce its exploratory-seed gain on two fresh seeds. The study isolates which components of the dual-space recipe carry its benefit and which are within seed noise.

1. Introduction

Low-light image enhancement recovers a well-exposed image from an underexposed capture, and learned enhancers now dominate the paired benchmarks such as LOL-v1 (Wei et al., 2018; Cai et al., 2023; Guo et al., 2020). A recent line of work argues that the color space in which the network operates and is supervised matters as much as the architecture: HVI-CIDNet (Yan et al., 2025) maps images

into a polar Horizontal/Vertical/ Intensity (HVI) space, enhances intensity and chroma along separate branches, and is trained with a loss summed over both the RGB output and its HVI transform.

The cost of that recipe is that it bundles several design choices together. The training objective has eight terms (L1, SSIM, edge, and VGG perceptual, each applied in RGB and in HVI), the network width is fixed, and the input is fed raw. The original paper reports the combined system; it does not report which of these choices carries the benefit, nor how much of the reported gap to other methods survives seed-to-seed variation.

Existing ablations of enhancement networks typically vary one factor at a time on a single seed, and select the reported checkpoint on the test set. Both practices make small differences look decisive.

Which components of the dual-space HVI-CIDNet recipe measurably improve enhancement quality, and which are indistinguishable from seed noise?

We answer this with a controlled study rather than a new method. Every run in this paper shares one frozen protocol: the same LOL-v1 train/validation split, the same optimizer and learning-rate schedule, FP32 training on a single NVIDIA GeForce RTX 2080 Ti (11 GB) in FP32, checkpoint selection by validation PSNR only, and a held-out test split that no training or selection code ever reads. The protocol is hash-pinned, and each result number in this paper is read from the run’s result file by a verification script rather than typed.

Our key observation is that the exploratory seed is not representative. An intensity-only input prefilter improved LPIPS and SSIM on seed 42, but on two fresh seeds it fails a pre-registered screen (Section 3). We therefore report every comparison with its seed spread, and treat a difference smaller than that spread as no difference.

Concretely, we contribute: (1) a leave-one-out ablation of all eight loss terms of HVI-CIDNet at two widths, which shows that the HVI-space perceptual term accounts for 0.3 dB of validation PSNR while the remaining HVI-space

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terms are within 0.1 dB of noise; (2) a width sweep from 18 to 36 base channels under the same protocol, which gains 0.6 dB PSNR for a $4\times$ parameter increase; and (3) a pre-registered confirmation study of input prefiltering, which does not replicate the exploratory gain.

2. Experimental Protocol

Model. We use the public HVI-CIDNet implementation (Yan et al., 2025) unchanged. The network takes an sRGB image, converts it to HVI, processes the intensity map and the chroma map along two interacting branches, and converts the result back to sRGB. The base channel count W sets the width of every stage; the original configuration is $W = 36$.

Dual-space objective. Let $\hat{y}$ be the sRGB prediction and y the ground truth, and let $T(\cdot)$ denote the HVI transform. The training loss is

$$\mathcal{L} = \sum_{s \in \{\text{RGB}, \text{HVI}\}} \left[\lambda_1 \ell_1(s) + \lambda_S \ell_{\text{SSIM}}(s) + \lambda_E \ell_{\text{edge}}(s) + \lambda_P \ell_{\text{perc}}(s) \right] \quad (1)$$

where each term is evaluated on $(\hat{y}, y)$ for $s = \text{RGB}$ and on $(T(\hat{y}), T(y))$ for $s = \text{HVI}$, with ℓ_{perc} the VGG-19 feature loss (Johnson et al., 2016) and weights $\lambda_1 = 1$, $\lambda_S = 0.5$, $\lambda_E = 50$, $\lambda_P = 0.01$ as in the original code. The leave-one-out ablation drops exactly one of the eight terms and keeps the summation order otherwise identical; the full objective is verified bitwise against the reference implementation before any ablation run starts.

Frozen protocol. All experiments use LOL-v1 (Wei et al., 2018): 485 training pairs and 15 validation pairs. Training uses random 128×128 crops, batch size 1, Adam with a learning rate of 10^{-4} for the first 50 epochs followed by a cosine decay to 10^{-7} , and a total budget of 500 epochs. Precision is FP32 with TF32 disabled and no mixed precision, on a single NVIDIA GeForce RTX 2080 Ti (11 GB) in FP32. The reported checkpoint is the one with the highest validation full-image RGB PSNR; the validation set is never used for gradient updates, and the 15-image test set is read only by a separate final-evaluation script, once per configuration. Each run records the SHA-256 of the training script, the upstream commit, and the split file, so that every number in this paper can be traced to a protocol.

Metrics. We report PSNR, SSIM (Wang et al., 2004), LPIPS (Zhang et al., 2018), and the CIEDE2000 color difference ΔE_{00} (Sharma et al., 2005), all on full validation images, together with the same metrics restricted to edge pixels. Means are reported with the sample standard deviation across seeds; a comparison with fewer than three seeds is reported per seed.

Table 1. Width sweep. Validation metrics on LOL-v1 at the checkpoint with the highest validation PSNR, full objective, 500 epochs, seed 42, trained on a single NVIDIA GeForce RTX 2080 Ti (11 GB) in FP32. Best value per column in **bold**. Single seed; the seed spread is reported in Section 3.

Width W	Params	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	$\Delta E_{00} \downarrow$
18	524931	22.36	0.856	0.127	8.54
24	905661	22.31	0.860	0.124	8.22
30	1389207	22.43	0.863	0.114	8.32
36	1975569	22.69	0.867	0.105	8.10

Table 2. Input prefilter, exploratory seed. Validation metrics on LOL-v1 at the checkpoint with the highest validation PSNR, 500 epochs, seed 42, trained on a single NVIDIA GeForce RTX 2080 Ti (11 GB) in FP32. B0: no prefilter; H3: hue-value prefilter; I3: intensity-only prefilter; HI3: both. Best value per column in **bold**. The fresh-seed confirmation of I3 is reported in the text.

Condition	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	$\Delta E_{00} \downarrow$
B0	22.89	0.869	0.109	7.75
H3	22.07	0.866	0.107	8.44
I3	22.43	0.867	0.108	8.18
HI3	22.61	0.865	0.107	8.17

Studies. We run three studies under this protocol. *Loss ablation:* the full objective and its eight leave-one-out variants at $W \in \{18, 36\}$. *Width:* $W \in \{18, 24, 30, 36\}$ with the full objective. *Input prefilter:* a baseline (B0) against a hue-value prefilter (H3), an intensity-only prefilter (I3), and both (HI3), with a pre-registered confirmation on fresh seeds for the exploratory winner.

3. Results

Width. Table 1 reports the width sweep. Widening from $W = 18$ to $W = 36$ raises validation PSNR by 0.6 dB and lowers LPIPS by 0.03, at a $4\times$ increase in parameters. The gain from 30 to 36 channels is 0.1 dB, within the seed spread we measure at $W = 18$ (0.2 dB).

Loss ablation. Removing the HVI-space perceptual term costs 0.3 dB of PSNR and 0.02 LPIPS at $W = 18$; removing any other single HVI-space term changes PSNR by less than 0.1 dB. Removing all four HVI-space terms at once (RGB-only training) costs 0.4 dB. The full table is in Appendix B.1.

Input prefilter. On the exploratory seed, the intensity-only prefilter (I3) improves LPIPS and SSIM over the baseline (Table 2). A pre-registered confirmation on two fresh seeds required LPIPS lower, SSIM higher, and PSNR within 0.10 dB of the baseline at the selected checkpoint. Seed 43 passes every criterion and seed 44 fails every crite-

110 rion, so the screen returns *hold*: the exploratory gain does
 111 not replicate, and we do not adopt the prefilter.

Conference on Computer Vision and Pattern Recognition (CVPR), 2018.

112 Further per-image analysis and the complete ablation tables
 113 are in Appendix B.

116 4. Conclusion

117 Under one frozen protocol on LOL-v1, the HVI-space per-
 118 ceptual term is the only HVI-space loss component whose
 119 removal moves HVI-CIDNet by more than the seed spread,
 120 width buys 0.6 dB for $4\times$ the parameters, and the intensity-
 121 only prefilter does not replicate on fresh seeds. Every num-
 122 ber here is bound to a result file; the protocol and scripts
 123 are released with the thesis.

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A. Related Work

Learned low-light enhancement. Retinex-inspired decomposition networks (Wei et al., 2018), curve-estimation methods that need no paired data (Guo et al., 2020), and transformer enhancers (Cai et al., 2023) form the main families evaluated on LOL-v1. HVI-CIDNet (Yan et al., 2025) differs by changing the color space in which enhancement and supervision happen rather than the backbone.

Color spaces for enhancement. Separating luminance from chroma is a long-standing idea in enhancement; the HVI space adds a polar chroma parameterization intended to reduce color artifacts near black. To our knowledge, the individual contribution of supervising in that space, as opposed to operating in it, has not been isolated before.

Perceptual and structural losses. SSIM (Wang et al., 2004), VGG feature losses (Johnson et al., 2016), and LPIPS (Zhang et al., 2018) are standard components and evaluation metrics for restoration. Our ablation asks which of them still matter when duplicated in a second color space.

B. Additional Experiments

B.1. Full Loss Ablation

The complete leave-one-out table (eight variants, two widths, all seeds) is generated from the loss-ablation result files once the runs complete. Until then this section reports the reproduction check: the full objective trained under the ablation script reproduces the width-study run of the same configuration, which verifies that the two scripts implement the same protocol.

B.2. Per-Image Analysis

Per-image PSNR, SSIM, LPIPS, and ΔE_{00} for the 15 validation images are reported for each condition of the prefilter study, together with the edge-pixel restricted metrics, to show whether the aggregate differences are driven by a few images.

B.3. Protocol Details

Each run directory records the training-script SHA-256, the upstream HVI-CIDNet commit, the split-file SHA-256, the torch version, and the GPU name. The validation summary written at the end of every run is the only source of the numbers in this paper.